Relational Data Model Exercises

1. How does the relational data model organize data?

A. As a tree of parent-child records
B. As tables consisting of tuples and attributes
C. As records connected by navigational links
D. As separate files whose relationships are defined only by applications

2. What typically establishes relationships between tables?

A. Matching table names
B. The positions of rows
C. Identical numbers of columns
D. Primary keys and foreign keys

3. What is a domain in the relational model?

A. A set of allowable atomic values for an attribute
B. The collection of all tables in a database
C. The number of attributes in a relation
D. A group of related records from different tables

4. A Salary domain permits values greater than 0 and less than 200,000. Which value belongs to this domain?

A. 0
B. −5,000
C. 75,000
D. 210,000

5. What does it mean for a value to be atomic?

A. It uniquely identifies a record.
B. It is treated as indivisible within the relational model.
C. It must be a single character or digit.
D. It cannot appear in more than one row.

6. Which pair of terms refers to a row and a column, respectively?

A. Attribute and tuple
B. Domain and relation
C. Relation and domain
D. Tuple and attribute

7. An Employee table contains 100 rows and five attributes. What is its degree?

A. 5
B. 20
C. 100
D. 500

8. What does it mean to say a relation is a subset of the Cartesian product of its domains?

A. Every possible combination of domain values must appear in the relation.
B. Each attribute must use the same domain.
C. Its tuples are selected combinations of values from the corresponding domains.
D. Each domain contains only values currently stored in the relation.

9. Which design violates the atomic-value property described in the slides?

A. A row containing separate EmployeeID, Name, and Salary values
B. A PhoneNumbers field containing a list of several phone numbers in one cell
C. Multiple employees having the same salary
D. A DepartmentID appearing in several employee records

10. Which statement describes a relation in the relational model?

A. Duplicate rows are allowed if they appear in different positions.
B. Row order determines which attribute each value belongs to.
C. Attribute names may repeat if their domains differ.
D. Rows are distinct, and the sequence of rows and columns is insignificant.